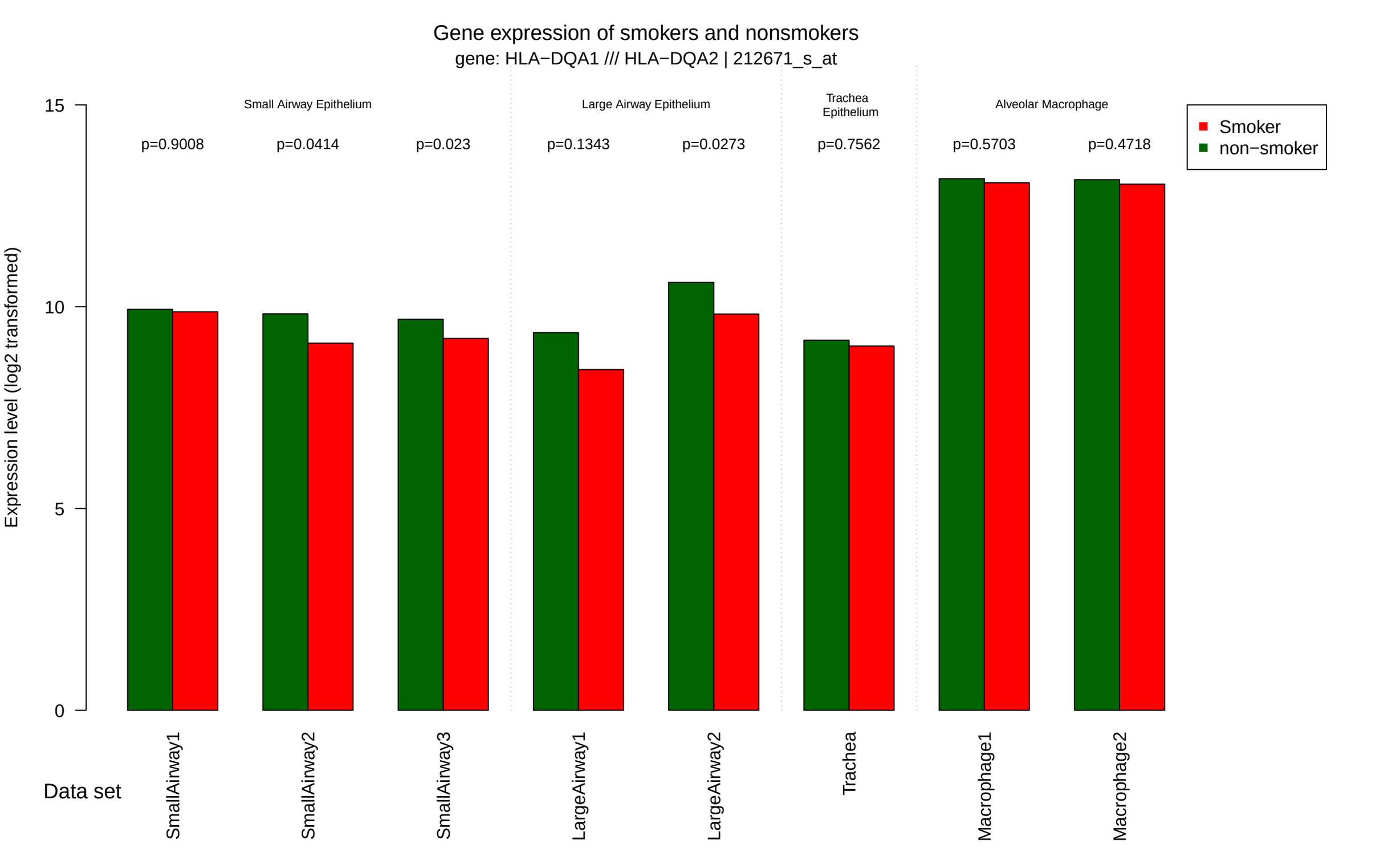
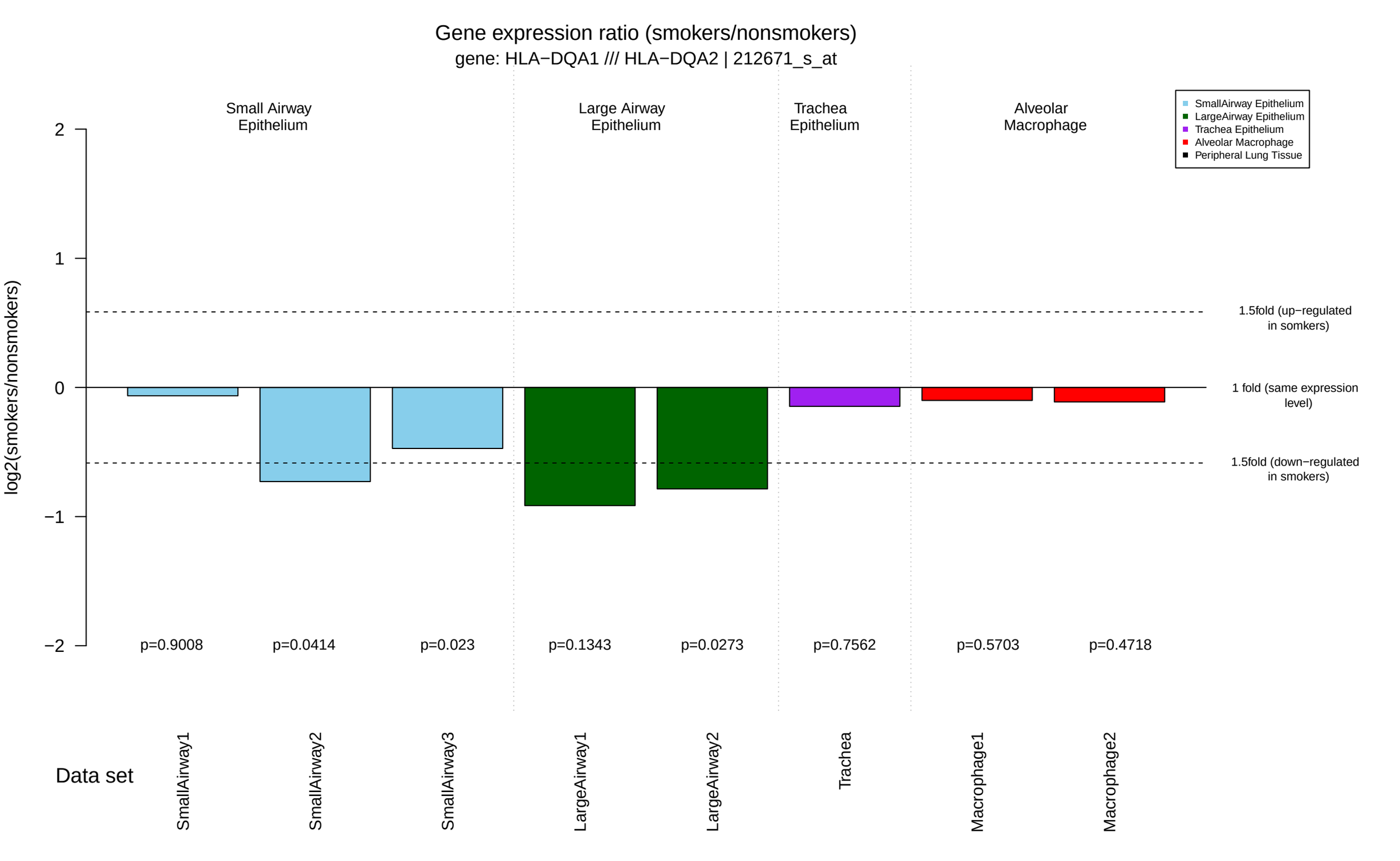
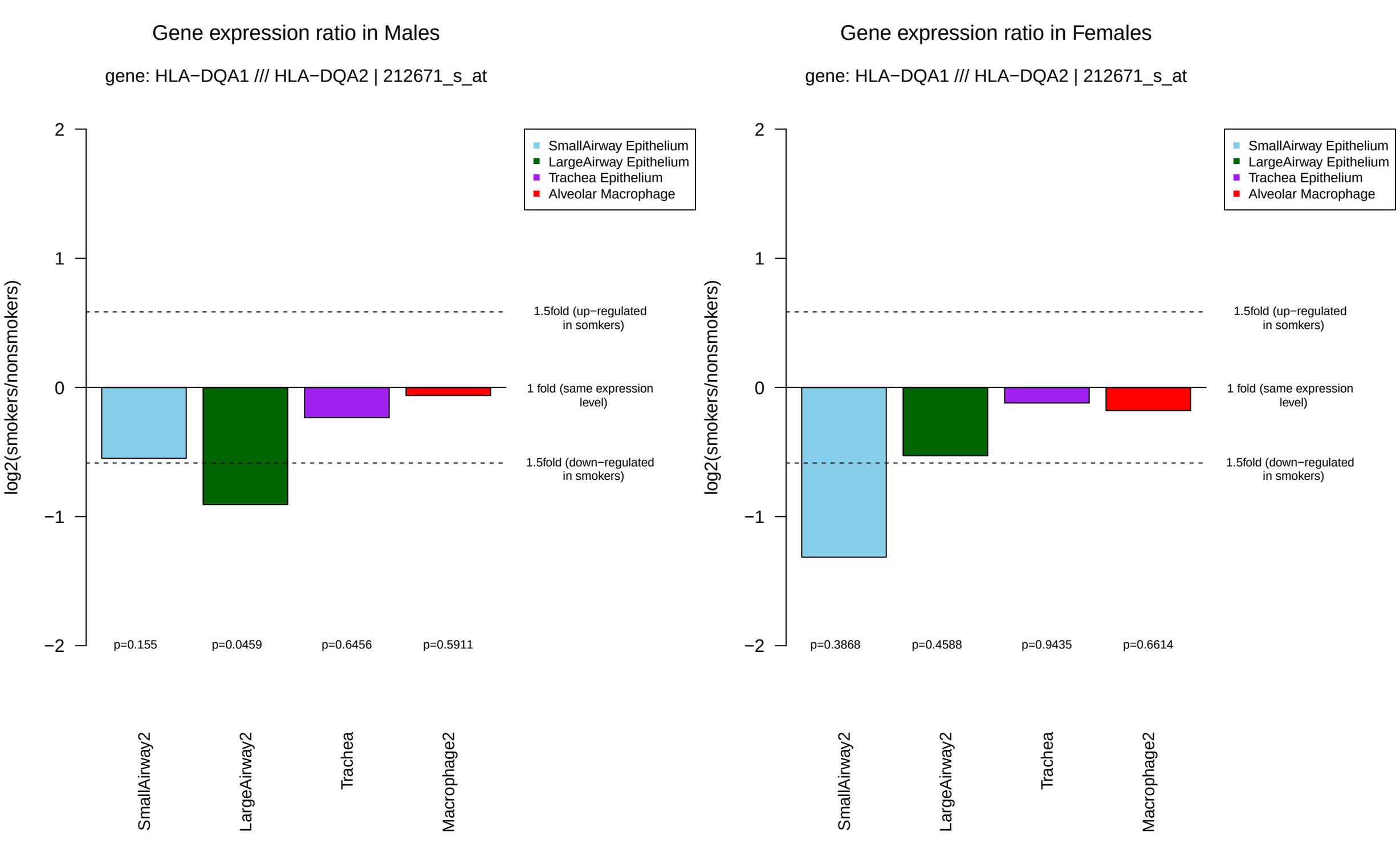
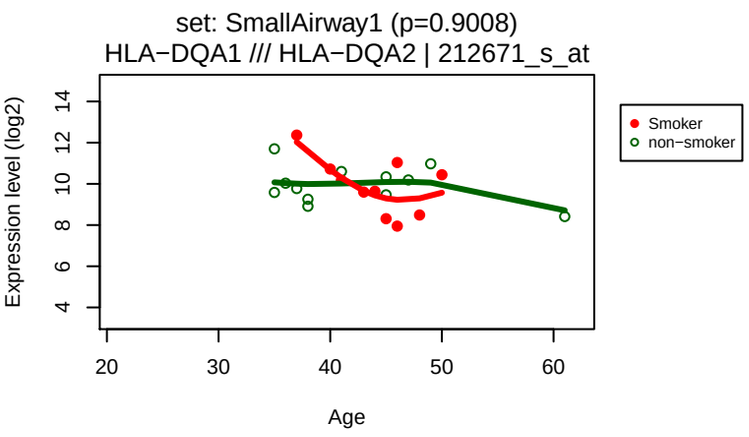
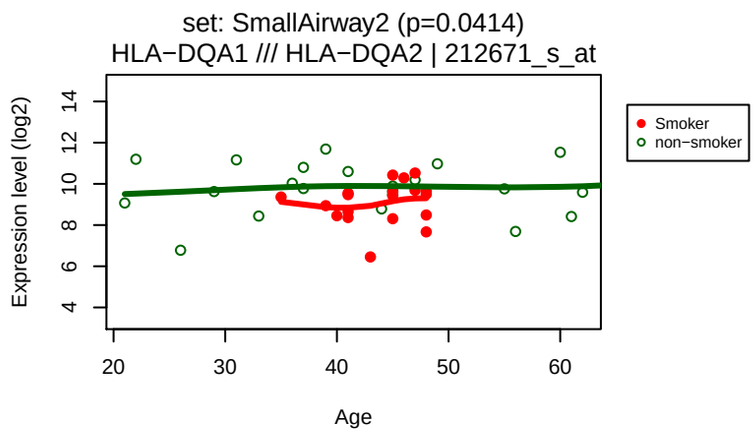
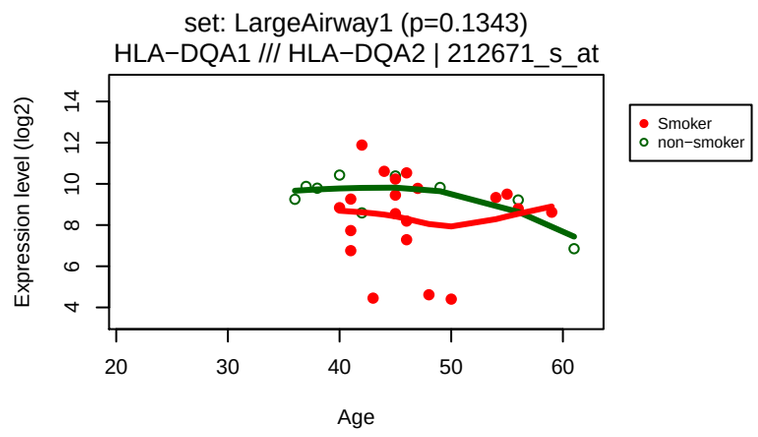

Gene expression of smokers and nonsmokers
gene: HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9008
p=0.0414
p=0.023
p=0.1343
p=0.0273
p=0.7562
p=0.5703
p=0.4718
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

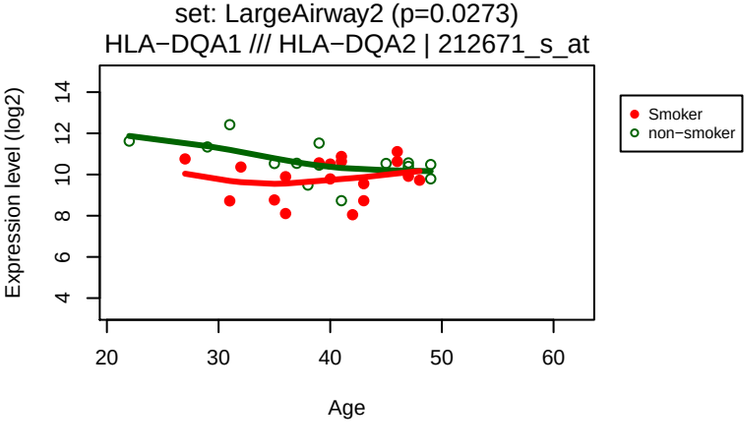

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9008
p=0.0414
p=0.023
p=0.1343
p=0.0273
p=0.7562
p=0.5703
p=0.4718
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

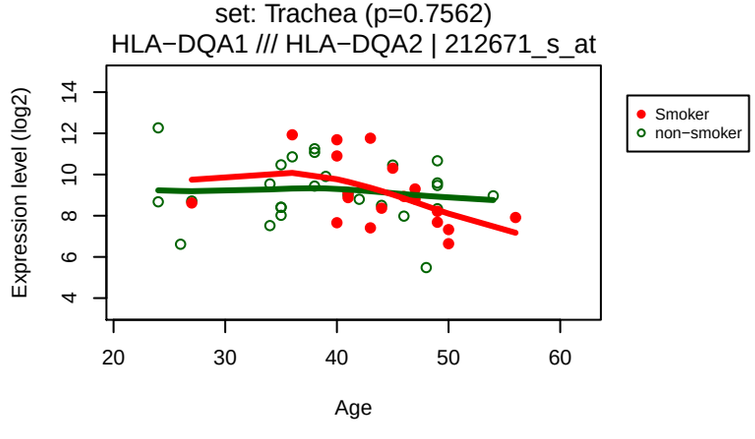

Gene expression ratio in Males
gene: HLA−DQA1 /// HLA−DQA2 | 212671_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.155
p=0.0459
p=0.6456
p=0.5911
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQA1 /// HLA−DQA2 | 212671_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3868
p=0.4588
p=0.9435
p=0.6614
SmallAirway2
LargeAirway2
Trachea
Macrophage2

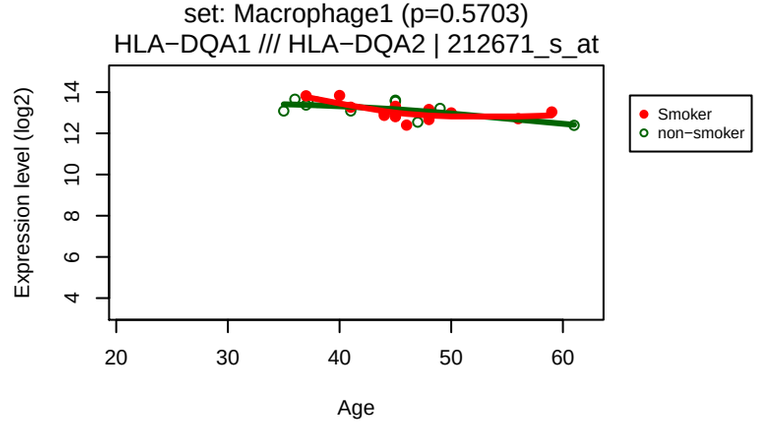

set: SmallAirway1 (p=0.9008)
HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Expression level (log2)
Age
Smoker
non−smoker

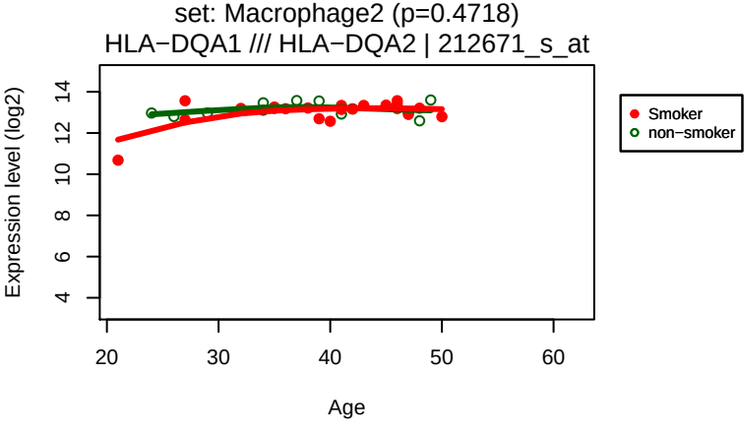

set: SmallAirway2 (p=0.0414)
HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Expression level (log2)
Age
Smoker
non−smoker

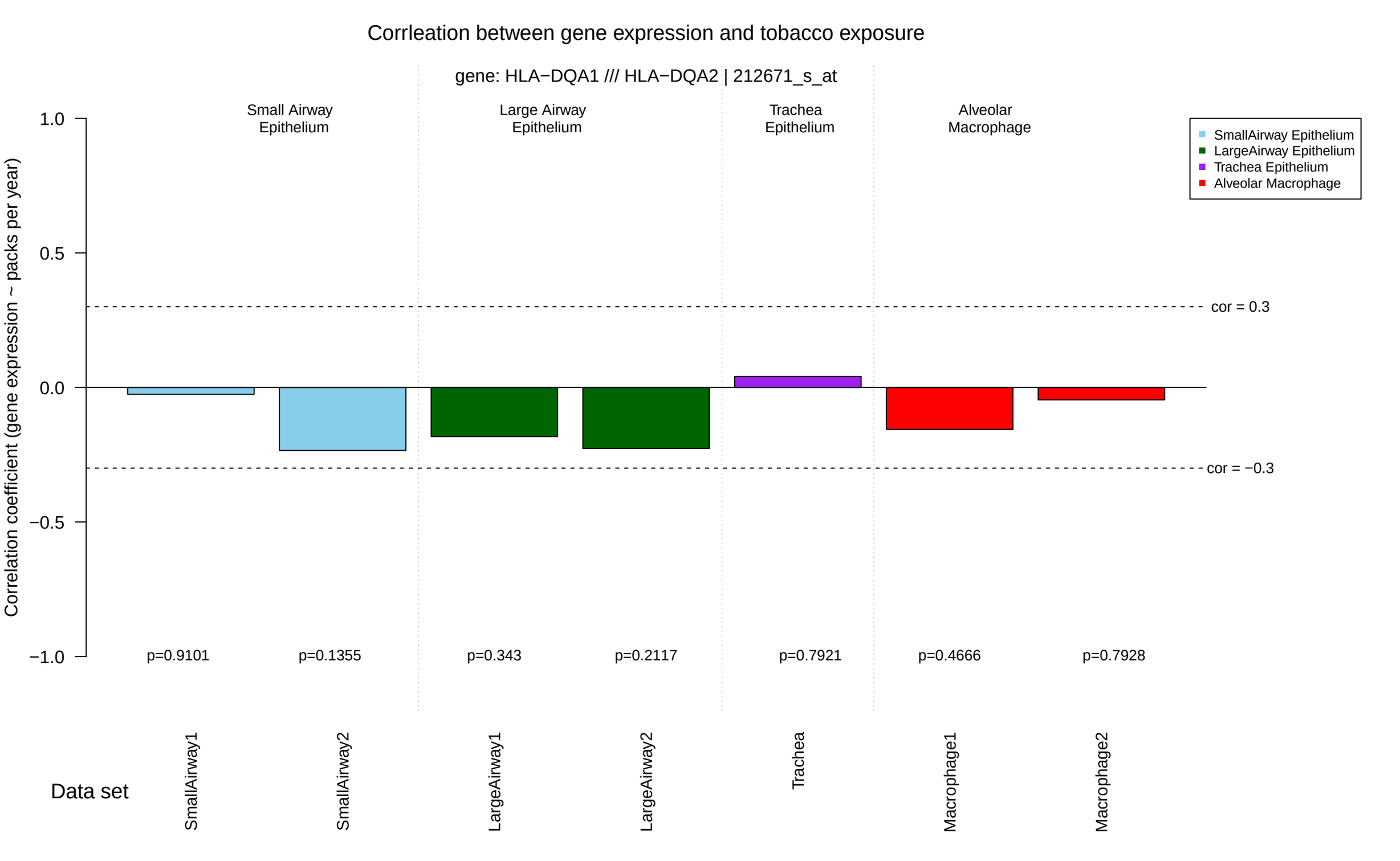

set: LargeAirway1 (p=0.1343)
HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0273)
HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7562)
HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5703)
HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4718)
HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQA1 /// HLA−DQA2 | 212671_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9101
p=0.1355
p=0.343
p=0.2117
p=0.7921
p=0.4666
p=0.7928
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2